

EE2683 – EMBEDDED SYSTEM AND IoT

PROJECT TITLE:

IoT BASED SMART IRRIGATION SYSTEM USING SOIL MOISTURE
SENSOR AND BLYNK

A MINI PROJECT REPORT

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ABSTRACT:

Agriculture is one of the most important sectors for human survival and economic development. Efficient water management is essential in agriculture because water scarcity and improper irrigation methods can negatively affect crop growth and productivity. Traditional irrigation systems require continuous manual supervision and often result in overwatering or underwatering of crops. These issues increase water wastage, labour costs, and energy consumption. To solve these problems, an automated Smart Irrigation System using a Soil Moisture Sensor and the Blynk application has been developed using Internet of Things (IoT) technology.

The proposed system is designed to monitor soil moisture levels continuously and provide automatic irrigation based on real-time environmental conditions. A soil moisture sensor is placed in the soil to detect the amount of water present. The sensor readings are processed by the Node MCU ESP8266 microcontroller, which acts as the main control unit of the system. The collected data is transmitted wirelessly to the Blynk mobile application through Wi-Fi technology. This enables farmers and users to monitor soil conditions and irrigation status remotely from anywhere using a smartphone.

When the soil moisture level drops below the predefined threshold value, the relay module automatically switches ON the water pump to irrigate the field. Once the soil reaches the desired moisture level, the relay turns OFF the pump automatically. This automation helps maintain proper soil moisture, ensuring healthy crop growth while avoiding unnecessary water usage. In addition to automatic operation, the system also allows users to manually control the motor through the mobile application whenever required.

The Smart Irrigation System offers several advantages such as water conservation, reduced human effort, improved irrigation efficiency, and lower electricity consumption. Real-time monitoring and wireless communication reduce the need for constant field inspection. The system also helps farmers save time and labour costs while improving agricultural productivity. Since the project uses low-cost components and simple circuitry, it is affordable and suitable for both small-scale and large-scale farming applications.

Another important feature of this project is its scalability and future enhancement capability. The system can be upgraded by integrating additional sensors such as temperature, humidity, rainfall, and water level sensors for more accurate environmental monitoring. Weather forecasting APIs and cloud storage can also be connected to analyse historical data and predict irrigation requirements. Furthermore, solar panels can be used as a power source to make the system energy-efficient and environmentally friendly.

This project demonstrates the practical application of IoT technology in modern agriculture and highlights the importance of smart farming techniques. By combining automation, wireless monitoring, and intelligent control, the Smart Irrigation System provides an effective solution for sustainable agriculture and efficient use.

INTRODUCTION:

Water is one of the most important resources in agriculture. In many places, farmers face problems due to irregular watering and water wastage. Traditional irrigation systems require manual labour and continuous monitoring. Improper irrigation can affect plant growth and crop yield.

The Internet of Things (IoT) plays an important role in modern agriculture by enabling smart monitoring and automation. Smart irrigation systems use sensors and internet connectivity to monitor soil conditions and automatically control watering systems. In this project, a soil moisture sensor measures the moisture level in the soil. The Node MCU ESP8266 microcontroller processes the sensor data and communicates with the Blynk cloud platform using Wi-Fi.

A relay module controls the water pump based on the soil condition. When the soil becomes dry, the pump turns ON automatically, and when sufficient moisture is detected, the pump turns OFF. This helps in maintaining proper soil moisture for healthy plant growth. The importance of this project lies in water conservation and efficient irrigation management. Water scarcity is becoming a major issue in many regions, making efficient use of water essential for agriculture.

The smart irrigation system reduces unnecessary water usage by supplying water only when needed. This improves water management, reduces wastage, and supports sustainable agricultural practices. Automation also reduces the need for continuous human supervision. This IoT-based irrigation system has many applications in agriculture and domestic environments. It can be used in farms, home gardens, greenhouses, nurseries, parks, and lawns.

The system is especially useful in areas where regular manual irrigation is difficult or where water resources are limited. Remote monitoring through the Blynk application allows users to check soil conditions and control the system from anywhere using a smartphone.

The significance of this project is that it provides a simple, low-cost, and efficient solution for modern irrigation problems. Proper irrigation improves crop quality, increases agricultural productivity, and supports healthy plant growth.

The use of IoT technology makes the system smart, reliable, and easy to operate. Therefore, the IoT-based smart irrigation system is an effective step toward smart farming and sustainable resource management.

Another important advantage of the smart irrigation system is the reduction of labour and energy consumption. In traditional irrigation methods, farmers need to manually check soil conditions and operate water pumps regularly. This process requires more time, effort, and manpower. By automating the irrigation process using IoT technology, the system can operate efficiently without constant human involvement. This not only saves labour costs but also reduces electricity consumption by operating the pump only when required. In addition, the project can be further improved by integrating advanced technologies such as temperature sensors, humidity sensors, rain detection sensors, and data analytics. These features can provide more accurate monitoring and better decision-making for irrigation management.

PROJECT SCOPE AND REQUIREMENTS:

The scope of the IoT-based smart irrigation system is to develop an automated watering system that monitors soil moisture and controls irrigation efficiently.

The project mainly focuses on reducing water wastage and minimizing manual effort in agricultural activities. It is designed for small farms, gardens, greenhouses, and agricultural lands where proper irrigation management is required.

The system ensures that water is supplied only when the soil moisture level falls below a specified limit.

The project uses IoT technology to provide real-time monitoring and automatic control of irrigation systems. The Node MCU ESP8266 microcontroller acts as the main processing unit and connects the system to the internet through Wi-Fi.

The Blynk cloud platform is used for remote monitoring and control through a smartphone application. This allows users to monitor soil conditions and pump status from any location with internet access.

The hardware requirements of the project include a Node MCU ESP8266 board, soil moisture sensor, relay module, water pump, connecting wires, power supply, and Wi-Fi connectivity.

The soil moisture sensor measures the moisture content in the soil and sends the data to the microcontroller. The relay module acts as a switching device to control the water pump automatically based on sensor readings. A stable power source is required for continuous operation of the system.

The software requirements include the Arduino IDE for programming the Node MCU ESP8266 and the Blynk application for IoT monitoring and control.

Embedded C programming is used to write the control logic for the system. Internet connectivity is necessary for communication between the microcontroller and the Blynk cloud platform. Proper configuration of the Blynk application and Wi-Fi settings is required for successful operation of the project.

The project scope can also be extended in the future by adding advanced sensors and technologies. Temperature sensors, humidity sensors, rainfall sensors, and GSM modules can be integrated to improve system performance and monitoring capabilities.

The system can also be expanded for large-scale agricultural applications using cloud data storage and data analysis techniques. Therefore, the project provides a flexible and scalable solution for modern smart farming applications.

The system can be further enhanced by integrating weather forecasting data to automatically adjust irrigation schedules based on rainfall prediction and environmental conditions. It can be expanded to support multiple soil moisture sensors in different field zones, enabling precision irrigation for large-scale farming with better water distribution control.

OBJECTIVES OF THE MINI PROJECT:

The primary objective of this project is to design and implement a smart irrigation system using Internet of Things (IoT) technology.

This system aims to modernize traditional irrigation methods by introducing automation and remote monitoring features, making the irrigation process more efficient and user-friendly.

Another important objective is to continuously monitor the soil moisture level in real time. By using a soil moisture sensor, the system is able to detect the exact water content in the soil, which helps in understanding the irrigation needs of the plants accurately and effectively.

The project also focuses on automatically controlling the water pump based on the detected soil moisture level. When the soil becomes dry, the system activates the pump, and when sufficient moisture is reached, the pump is turned off. This automation ensures proper water supply without the need for manual intervention.

In addition, one of the key objectives is to reduce water wastage and minimize manual effort.

By supplying water only when required, the system helps in conserving water resources and reducing unnecessary usage, which is especially important in areas facing water scarcity.

Finally, the system enables users to monitor and control the irrigation process remotely using the Blynk mobile application. This feature provides convenience and flexibility, allowing users to manage irrigation from anywhere using their smartphones.

This project also aims to improve the efficiency and reliability of irrigation systems by minimizing human errors. Manual irrigation often depends on estimation, which can lead to improper watering. By introducing an automated system, the accuracy of water supply is significantly improved.

Another objective is to promote smart agriculture practices by integrating IoT technology into farming. This helps in modernizing traditional agricultural methods and encourages farmers to adopt technology-driven solutions for better productivity and resource management.

The system is also designed to be cost-effective and easily implementable using commonly available components such as microcontrollers and sensors.

This ensures that even small-scale farmers and home gardeners can benefit from smart irrigation without high investment costs.

Furthermore, the project aims to support sustainable agricultural development by conserving water resources and maintaining optimal soil conditions.

This contributes to environmental protection while ensuring healthy plant growth and improved crop yield.

SYSTEM DESIGN:

BLOCK DIAGRAM:

The below Figure 1.0 shows the flow of the Smart Irrigation System using Soil Moisture Sensor, Node MCU ESP8266, Wi-Fi, Blynk App, Relay Module, and Water Pump.

Soil Moisture Sensor:

The soil moisture sensor is used to measure the moisture content present in the soil. It detects whether the soil is dry or wet by sensing the water level around the plant roots.

The sensor sends analog or digital signals to the Node MCU ESP8266 based on the soil condition. If the soil moisture level is low, the system identifies that watering is required. This sensor helps in maintaining proper irrigation and prevents overwatering or underwatering of plants.

Node MCU ESP8266:

The Node MCU ESP8266 is the main microcontroller used in the project. It processes the data received from the soil moisture sensor and controls the entire irrigation system.

The Node MCU has built-in Wi-Fi capability, which allows it to connect to the internet and communicate with the Blynk cloud platform. Based on the sensor readings, the microcontroller decides whether the water pump should be turned ON or OFF. It acts as the brain of the smart irrigation system.

Wi-Fi:

Wi-Fi is used for wireless communication between the Node MCU ESP8266 and the Blynk cloud platform. It enables the system to send sensor data to the internet and receive control commands remotely.

Through Wi-Fi connectivity, users can monitor the irrigation system using a smartphone from any location. This wireless communication makes the system smart, flexible, and suitable for IoT-based applications.

Blynk App:

The Blynk application is an IoT platform used for monitoring and controlling the irrigation system through a smartphone. It displays real-time soil moisture values and pump status to the user. The app allows users to remotely monitor the condition of the soil and manually control the water pump if needed. Blynk provides an easy-to-use graphical interface, making the system more user-friendly and convenient for farmers and users.

Relay Module:

The relay module acts as an electrically operated switch in the irrigation system. It receives control signals from the Node MCU ESP8266 and controls the operation of the water pump. Since the water pump operates at a

higher voltage than the microcontroller, the relay provides electrical isolation and safe switching. When the soil becomes dry, the relay turns the pump ON, and when sufficient moisture is detected, it turns the pump OFF automatically.

Water Pump:

The water pump is responsible for supplying water to the plants or agricultural field. It receives power through the relay module and operates automatically based on soil moisture conditions. When the soil is dry, the pump delivers water to maintain proper moisture levels. Once the required moisture is reached, the pump stops automatically.

This helps in efficient irrigation, water conservation, and healthy plant growth.

The basic system design of the smart irrigation system follows a simple IoT-based flow starting from the soil moisture sensor, which continuously measures the moisture level of the soil.

This data is then sent to the Node MCU ESP8266 microcontroller, which acts as the main processing unit of the system.

The Node MCU analyses the sensor readings and connects to the internet through Wi-Fi to enable remote communication.

The processed data is then transmitted to the Blynk application, where the user can monitor the real-time soil moisture level on a smartphone.

Based on the moisture condition, the Node MCU also controls a relay module, which acts as a switch for the water pump.

When the soil is dry, the relay activates the pump to supply water, and when the required moisture level is reached, the pump is automatically turned off, ensuring efficient and automated irrigation control

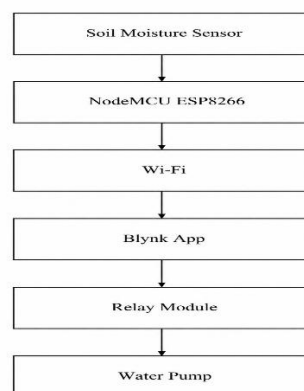


FIGURE 1.0

SYSTEM REQUIREMENTS:

HARDWARE REQUIREMENTS:

Table 1.0 shows the list of hardware components used in the Smart Irrigation System project.

1. Node MCU ESP8266

A Wi-Fi enabled microcontroller used to control the entire irrigation system. It reads sensor data and automates the water pump operation.

2. Soil Moisture Sensor

This sensor detects the moisture level present in the soil.

3. Relay Module

The relay module acts as a switch for controlling high-power devices. It safely turns the water pump ON and OFF using Node MCU signals.

4. Water Pump

The water pump supplies water to the plants automatically. It starts working whenever the soil becomes dry.

5. Jumper Wires

Jumper wires are used to connect electronic components together.

6. Breadboard

A breadboard is used for building circuits without soldering.

S. No	Components
1	Node MCU ESP8266
2	Soil Moisture Sensor
3	Relay Module
4	Water Pump
5	Jumper Wires
6	Breadboard
7	Battery / Power Supply

Table 1.0

SOFTWARE REQUIREMENTS:

The smart irrigation system is developed using various software tools and libraries that support programming, connectivity, and remote monitoring.

These software components play a crucial role in enabling communication between the hardware devices and the IoT platform, ensuring smooth functioning of the entire system. Table 1.1 shows the software tools and libraries used in the Smart Irrigation System project.

1.ArduinoIDE

Arduino IDE is the primary software used for writing, compiling, and uploading code to the Node MCU ESP8266 microcontroller. It provides a simple and user-friendly interface for developing embedded system applications. The IDE supports C/C++ programming language and includes built-in functions that make it easier to control sensors, modules, and other hardware components used in the project. It is also useful for debugging and testing the code during development.

2.BlynkIoTApplication

The Blynk IoT application is used for real-time monitoring and remote control of the irrigation system. It allows users to view soil moisture data on a smartphone and control the water pump from anywhere using an internet connection. The application provides a graphical interface with widgets such as gauges, buttons, and displays, making it easy for users to interact with the system. This enhances convenience and enables smart farming practices.

3.ESP8266BoardPackage

The ESP8266 board package is required to integrate the Node MCU ESP8266 with the Arduino IDE. It includes the necessary drivers, board definitions, and configuration settings needed for programming the microcontroller. By installing this package, the Arduino IDE can properly recognize and communicate with the Node MCU board, allowing developers to upload and execute code without compatibility issues.

4.BlynkLibrary

The Blynk Library is an important software component that enables communication between the Node MCU and the Blynk IoT platform. It allows the system to send sensor data, such as soil moisture readings, to the mobile application and receive control commands from the user. The library manages Wi-Fi connectivity and ensures smooth data transfer between hardware and software, making the system efficient and reliable.

S. No	Software
1	Arduino IDE
2	Blynk IoT Application
3	ESP8266 Board Package
4	Blynk Library

Table 1.1

WORKING PRINCIPLE:

The Soil Moisture Sensor continuously monitors the water content present in the soil. It detects the moisture condition of the soil and helps determine whether the soil is dry, normal, or wet based on the measured moisture levels. This enables the system to understand the irrigation requirement of the plants in real time.

Once the soil condition is detected, the sensor converts the moisture data into an electrical signals. These signals represent the variation in soil moisture and are suitable for further processing by the microcontroller. The accuracy of these signals plays an important role in ensuring proper irrigation control.

The analog signals from the sensor are then sent to the Node MCU ESP8266 for processing. The Node MCU acts as the main controller of the system and compares the received moisture values with a predefined threshold level. Based on this comparison, it decides whether irrigation is required or not.

The processed data is also transmitted to the Blynk IoT application through Wi-Fi connectivity. This allows users to monitor real-time soil moisture readings directly from the Blynk mobile dashboard. It provides remote access and improves user convenience in managing the irrigation system.

When the moisture value falls below the set threshold, the Node MCU automatically activates the relay module. The relay then switches ON the water pump, supplying water to the plants. Water continues to flow until the soil reaches the required moisture level.

Once sufficient moisture is detected, the Node MCU deactivates the relay module, which turns OFF the water pump. This prevents overwatering and helps conserve water resources. The system continuously repeats this monitoring and control process, ensuring efficient and automated irrigation management. Additionally, users can manually control the water pump through the Blynk application whenever required for greater flexibility.

The system also improves overall water management by ensuring that water is supplied only when necessary. This helps in reducing unnecessary water usage and supports conservation of water resources, which is especially important in areas facing water scarcity. By automating the irrigation process, the system eliminates human errors that may occur during manual watering.

Another advantage of this system is its ability to provide continuous and real-time monitoring of soil conditions. Farmers or users do not need to physically check the soil every time, as all updates are available on the Blynk application. This saves time and effort while improving decision-making in irrigation control.

In addition, the system enhances agricultural productivity by maintaining optimal soil moisture levels for plant growth. Proper irrigation ensures healthier crops and better yield quality. The integration of IoT technology makes the system more efficient, reliable, and suitable for modern smart farming applications.

ARDUINO PROGRAM:

```
#define BLYNK_TEMPLATE_ID "YourTemplateID"
#define BLYNK_TEMPLATE_NAME "Smart Irrigation"
#define BLYNK_AUTH_TOKEN "YourAuthToken"
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
char ssid[] = "WiFi_Name";
char pass[] = "WiFi_Password";
#define sensorPin A0
#define relayPin D1
BlynkTimer timer;
void sendSensor()
{
  int moisture = analogRead(sensorPin);
  Serial.print("Moisture Value: ");
  Serial.println(moisture);
  Blynk.virtualWrite(V0, moisture);
  if(moisture > 700)
  {
    digitalWrite(relayPin, LOW);
    Blynk.virtualWrite(V1, 1);
  }
  else
  {
    digitalWrite(relayPin, HIGH);
    Blynk.virtualWrite(V1, 0);
  }
}
BLYNK_WRITE(V1)
{
  int value = param.asInt();
  if(value == 1)
  {
    digitalWrite(relayPin, LOW);
  }
  else
  {
    digitalWrite(relayPin, HIGH);
  }
}
void setup()
{
  Serial.begin(9600);
  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, HIGH);
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  timer.setInterval(2000L, sendSensor);
}
void loop()
{
  Blynk.run();
  timer.run();
}
```

EXPLANATION OF CODE:

The program begins by initializing the Node MCU ESP8266, Wi-Fi connection, and Blynk IoT application authentication settings during startup.

This setup ensures that the microcontroller is properly connected to the internet and ready to communicate with the Blynk platform for remote monitoring and control of the irrigation system.

After initialization, the code continuously reads analog values from the soil moisture sensor. These values represent the moisture content present in the soil and are used to determine whether the soil is dry, normal, or wet. The sensor readings are essential for making accurate irrigation decisions.

The collected sensor data is then processed and converted into readable moisture percentage values. This conversion makes it easier for users to understand the soil condition. The processed data is also sent to the Blynk application, where it can be viewed in real time through a mobile dashboard.

The program then compares the current moisture value with a predefined threshold value set in the code. If the soil moisture level falls below this threshold, the Node MCU sends a signal to the relay module.

The relay then activates the water pump, starting the irrigation process automatically.

When the soil moisture reaches the required level, the system automatically switches OFF the water pump by deactivating the relay module. This ensures that plants receive the correct amount of water without over-irrigation, helping to conserve water resources effectively.

The code continuously updates the sensor readings in a loop to maintain real-time monitoring and efficient irrigation control. Additionally, the system includes manual control features through the Blynk application, allowing users to turn the pump ON or OFF remotely whenever needed for better flexibility and control.

The system improves reliability by ensuring that irrigation decisions are made based on real-time sensor data rather than manual estimation. This reduces the chances of human error and ensures that plants receive water only when it is actually required, leading to more accurate and efficient irrigation management.

Another important feature of the program is its ability to maintain continuous communication between the Node MCU and the Blynk server.

This stable connection allows uninterrupted data transfer, ensuring that users always receive updated soil moisture readings and system status on their mobile devices.

Overall, the programming logic supports automation, remote access, and efficient water usage in agriculture. By combining sensor-based monitoring with IoT control, the system provides a smart and sustainable solution for modern irrigation needs, making farming more productive and technology-driven.

CLOUD PLATFORM EXPLANATION:

The Blynk IoT Application is used as the cloud platform for monitoring and controlling the smart irrigation system. It provides a user-friendly dashboard that connects the Node MCU ESP8266 to the internet through Wi-Fi. The platform allows real-time communication between the hardware components and the mobile application. This enables users to access the irrigation system remotely from anywhere.

The Blynk dashboard displays important information such as soil moisture readings, pump status, and system activity in real time. Data collected from the Soil Moisture Sensor is sent by the Node MCU to the Blynk cloud server using internet connectivity. The cloud platform processes and updates this information instantly on the mobile dashboard. Users can easily monitor whether the soil is dry or wet without physically checking the field.

The Blynk platform also provides manual control features for the irrigation system. Through virtual buttons and switches available on the dashboard, users can manually turn the Water Pump ON or OFF whenever required. This feature is useful during maintenance, emergency situations, or when users want direct control over irrigation. Notifications and alerts can also be configured to inform users about soil conditions or pump activity.

Using the Blynk cloud platform improves automation, remote accessibility, and system efficiency. It eliminates the need for constant human supervision and supports smart farming practices. The dashboard interface is simple, interactive, and easy to use, making the irrigation system more reliable and convenient for users.

The Blynk IoT Application also ensures secure and stable communication between the hardware and the user interface. It uses cloud-based servers to manage data transmission, which helps in maintaining reliable connectivity even when the user is not physically near the system. This makes the irrigation setup more dependable for long-term agricultural use.

Another important advantage of the Blynk cloud platform is its ability to store and visualize historical data. Users can analyse past soil moisture trends and pump activity, which helps in understanding irrigation patterns. This data can be useful for improving farming decisions and optimizing water usage over time.

The platform also supports customization of the mobile dashboard according to user requirements. Widgets such as gauges, graphs, and notification panels can be arranged easily to display relevant information. This flexibility allows users to design a control panel that best suits their monitoring and control needs.

In addition, the Blynk system helps in reducing the dependency on physical presence in the field. Farmers can monitor and control the irrigation system from any location using a smartphone with internet access. This remote accessibility is especially beneficial for large farms or areas that are difficult to visit frequently.

Overall, the integration of the Blynk cloud platform enhances the performance and intelligence of the irrigation system. It combines automation, real-time monitoring, and remote control into a single interface, making the system more efficient, user-friendly, and suitable for modern smart agriculture applications.

SENSOR INTEGRATION:

The **Soil Moisture Sensor** plays an important role in the smart irrigation system by detecting the moisture content present in the soil. It is inserted into the soil near the plant roots, where it continuously monitors the water level. The sensor works by measuring the conductivity between its probes, which changes according to the amount of moisture in the soil. The collected data is then converted into electrical signals for further processing.

The sensor is connected to the analog input pin of the **Node MCU ESP8266**. The Node MCU continuously reads the analog values sent by the sensor and compares them with a predefined threshold value programmed in the code. If the moisture level falls below the threshold, the system identifies that the soil is dry and requires watering. This automated sensing process helps maintain proper soil conditions without human intervention.

The **Relay Module** is integrated into the system to control the operation of the **Water Pump**. Since the Node MCU cannot directly handle high voltage or high current devices, the relay acts as an intermediate switching device. When the Node MCU detects dry soil conditions, it sends a control signal to the relay module, which then switches ON the water pump. The pump supplies water to the plants automatically until sufficient moisture is restored in the soil.

Once the soil reaches the required moisture level, the sensor sends updated readings to the Node MCU. The controller then deactivates the relay module, which turns OFF the water pump and stops irrigation. This complete integration of the soil moisture sensor and relay module ensures efficient water usage, reduces manual effort, and supports automatic irrigation management. The system also improves plant health by providing water only when necessary.

The sensor integration in this system also ensures high accuracy in detecting soil conditions. Since the Soil Moisture Sensor continuously measures real-time moisture levels, it helps in avoiding incorrect irrigation decisions. This improves the reliability of the entire irrigation system and ensures that plants receive the correct amount of water based on actual soil requirements.

Another important aspect of integration is the proper calibration of sensor values. The analog readings from the sensor are mapped into meaningful moisture levels using threshold limits in the program. This calibration process is essential to differentiate between dry, normal, and wet soil conditions, making the system more precise and adaptable to different soil types.

The integration between the sensor, Node MCU, and relay module ensures smooth and efficient communication within the system. The Node MCU acts as the central controller that processes sensor data and coordinates actions between input and output devices. This seamless interaction allows the system to function automatically without delay or manual intervention.

Overall, the sensor integration enhances the performance and intelligence of the smart irrigation system. By combining real-time sensing, automatic control, and safe switching mechanisms, the system achieves efficient water management.

RESULTS:

The Smart Irrigation System was successfully implemented and tested under different soil conditions. The system operated as expected, showing reliable performance in detecting soil moisture levels and controlling the irrigation process automatically. The integration of all components worked efficiently without errors during testing.

The Soil Moisture Sensor accurately measured the moisture content present in the soil and continuously sent real-time data to the Node MCU ESP8266 for processing. The readings were stable and responsive to changes in soil conditions, ensuring proper monitoring of plant water requirements.

Based on the processed sensor values, the system automatically controlled the water pump using the relay module. When the soil was dry, the pump turned ON, and when sufficient moisture was detected, it turned OFF. This automatic switching ensured efficient irrigation without manual intervention.

The Blynk IoT application successfully displayed real-time soil moisture data and system status on the mobile dashboard. It also allowed users to monitor and control the irrigation system remotely, improving convenience and accessibility. The communication between hardware and cloud platform was smooth and consistent.

OBSERVATIONS:

During testing, it was observed that the system responded correctly to changes in soil conditions. The sensor readings varied depending on the moisture content, allowing accurate detection of dry and wet soil states. The threshold-based control worked effectively in triggering the irrigation process.

When the soil condition was dry, with moisture values above 700, the system automatically turned the pump ON to supply water. This ensured that plants received adequate irrigation whenever the soil lacked sufficient moisture.

When the soil condition was wet, with moisture values below 700, the system automatically turned the pump OFF. This helped prevent overwatering and ensured proper conservation of water resources.

Overall, the observations confirmed that the system operates reliably under different conditions, maintaining efficient irrigation control and consistent performance throughout the testing phase. Table 1.2 shows the relationship between soil moisture condition, moisture sensor values, and water pump status in the Smart Irrigation System.

Soil Condition	Moisture Value	Pump Status
Dry Soil	Above 700	ON
Wet Soil	Below 700	OFF

Table 1.2

OUTPUT FEATURES:

The Smart Irrigation System provides real-time soil moisture monitoring, allowing users to continuously observe the current condition of the soil.

This helps in making timely irrigation decisions based on actual data rather than assumptions.

One of the major features of the system is automatic water pump control.

The system intelligently switches the motor ON or OFF depending on the moisture level detected by the sensor, ensuring smooth irrigation without human intervention.

The system also offers remote monitoring through the Blynk mobile application.

Users can easily check soil moisture levels and system status from anywhere using a smartphone with internet connectivity.

Another important feature is the reduction of water wastage.

Since irrigation is carried out only when the soil is dry, unnecessary water usage is avoided, supporting efficient water conservation.

The system is designed for easy operation, making it suitable even for users without technical knowledge. Once installed, it functions automatically and requires very little manual supervision.

It is also a low-cost solution, as it uses affordable components like Node MCU ESP8266, soil moisture sensor, relay module, and water pump. This makes it accessible for small-scale farmers and home gardeners.

The system improves agricultural productivity by maintaining optimal soil moisture levels.

Proper irrigation ensures healthier plant growth and enhances overall crop yield quality.

Another key feature is its real-time data display on the Blynk dashboard.

Users can view updated moisture readings, pump status, and system activity instantly on their mobile devices.

The system also supports flexible control options, allowing users to manually override the automatic mode using the Blynk application whenever required.

This adds convenience and better control over irrigation.

Finally, the integration of IoT technology makes the system scalable and future-ready.

APPLICATIONS:

The Smart Irrigation System is widely used in agriculture fields to automate watering based on real-time soil moisture conditions.

It helps farmers maintain proper irrigation without continuous manual monitoring. In home gardens, it ensures plants receive the right amount of water even when the owner is not present.

It is also useful in greenhouses where controlled moisture levels are required for healthy plant growth. Additionally, it can be applied in parks, plantations, and landscaping areas for efficient water management.

This system is also suitable for modern smart farming and IoT-based agricultural projects. It supports automated irrigation in large farming areas where manual watering is difficult and time-consuming.

By integrating sensors and wireless communication, it enables remote monitoring of soil conditions. It improves agricultural productivity by ensuring plants receive water at the right time.

Overall, it is an important solution for efficient and technology-based farming systems.

ADVANTAGES:

The system helps in saving water by irrigating only when the soil is dry, preventing unnecessary water usage. It reduces human effort by automating the entire irrigation process

The use of **Soil Moisture Sensor** and **Node MCU ESP8266** ensures accurate detection and control of soil moisture levels. It also provides real-time monitoring and control through the **Blynk IoT Application**, making the system more user-friendly and efficient.

The system improves crop health by maintaining optimal moisture levels in the soil at all times. It reduces wastage of resources such as water and electricity, making it cost-effective.

The automation feature minimizes the need for manual supervision and increases efficiency in irrigation management. It is reliable, easy to use, and suitable for both small-scale and large-scale agricultural applications.

Overall, it supports smart farming and sustainable agriculture practices.

REAL TIME USAGE:

The system provides instant response to changes in soil moisture levels, ensuring that irrigation decisions are made immediately without any delay. This real-time reaction helps maintain proper soil conditions and prevents plant stress. It enables continuous 24/7 monitoring of the irrigation system, allowing users to track soil conditions and pump status at any time.

CHALLENGES AND ISSUES FACED:

During the development of the Smart Irrigation System, several challenges and technical issues were encountered while integrating the hardware and software components. One of the major difficulties was obtaining accurate readings from the **Soil Moisture Sensor**.

The sensor readings sometimes varied due to changes in soil type, temperature, and water content. Calibration was required multiple times to set the correct threshold value for detecting dry and wet soil conditions accurately.

Another challenge was maintaining a stable Wi-Fi connection with the **Node MCU ESP8266**. Since the system depends on internet connectivity for communication with the **Blynk IoT Application**, weak or unstable Wi-Fi signals occasionally interrupted data transmission.

This caused delays in updating moisture values on the dashboard and affected remote monitoring performance. Reconnecting the device and optimizing the network settings helped improve communication reliability.

Power management was also an important issue during the implementation process. The **Water Pump** required more current than the Node MCU could directly provide.

To solve this issue, a **Relay Module** was used as an intermediate switching device. However, improper wiring and insufficient power supply initially caused the relay to malfunction. Additional care was taken to ensure proper voltage levels and secure electrical connections.

Integrating the hardware components with the software environment was another challenge faced during the project. Installing the **ESP8266 Board Package** and configuring the **Blynk Library** in the **Arduino IDE** required careful setup.

Errors in coding, incorrect pin configurations, and library compatibility issues occasionally caused the program to fail during testing. These problems were solved through debugging and repeated testing of the code.

Environmental factors also affected the overall performance of the irrigation system. Moisture sensors exposed to water for long durations experienced corrosion, which reduced sensor efficiency over time.

Additionally, fluctuations in power supply sometimes caused the Node MCU to restart automatically. Proper insulation, stable power sources, and protective casing were considered to improve the durability and reliability of the system.

Despite these challenges, all issues were gradually resolved through troubleshooting, testing, and optimization.

The project provided valuable practical knowledge in IoT, sensor integration, cloud connectivity, and automation systems.

CONNECTION DIAGRAM:

This connection diagram shows how an Esp. If Systems ESP8266 Wi-Fi module controls a mini submersible water pump using a 1-channel relay board. The ESP8266 acts as the main controller, while the relay works as an electronic switch to turn the pump ON and OFF. Such circuits are commonly used in IoT-based watering and automation systems.

The ESP8266 is powered with 3.3 V, and its GPIO2 pin is connected to the relay input pin. The relay module uses a separate 5 V supply and shares a common ground with the ESP8266 for proper communication. When the ESP8266 sends a signal through GPIO2, the relay activates and switches the connected pump circuit. Figure 1.1 shows the circuit connection of the Smart Irrigation System using ESP8266 Wi-Fi module, relay module, and mini water pump.

The water pump is connected through the relay terminals so that the relay can safely control the higher-power load. When the relay is ON, current flows to the pump and it starts operating; when OFF, the pump stops. This setup protects the ESP8266 from high current and allows wireless control of the pump through Wi-Fi programming.

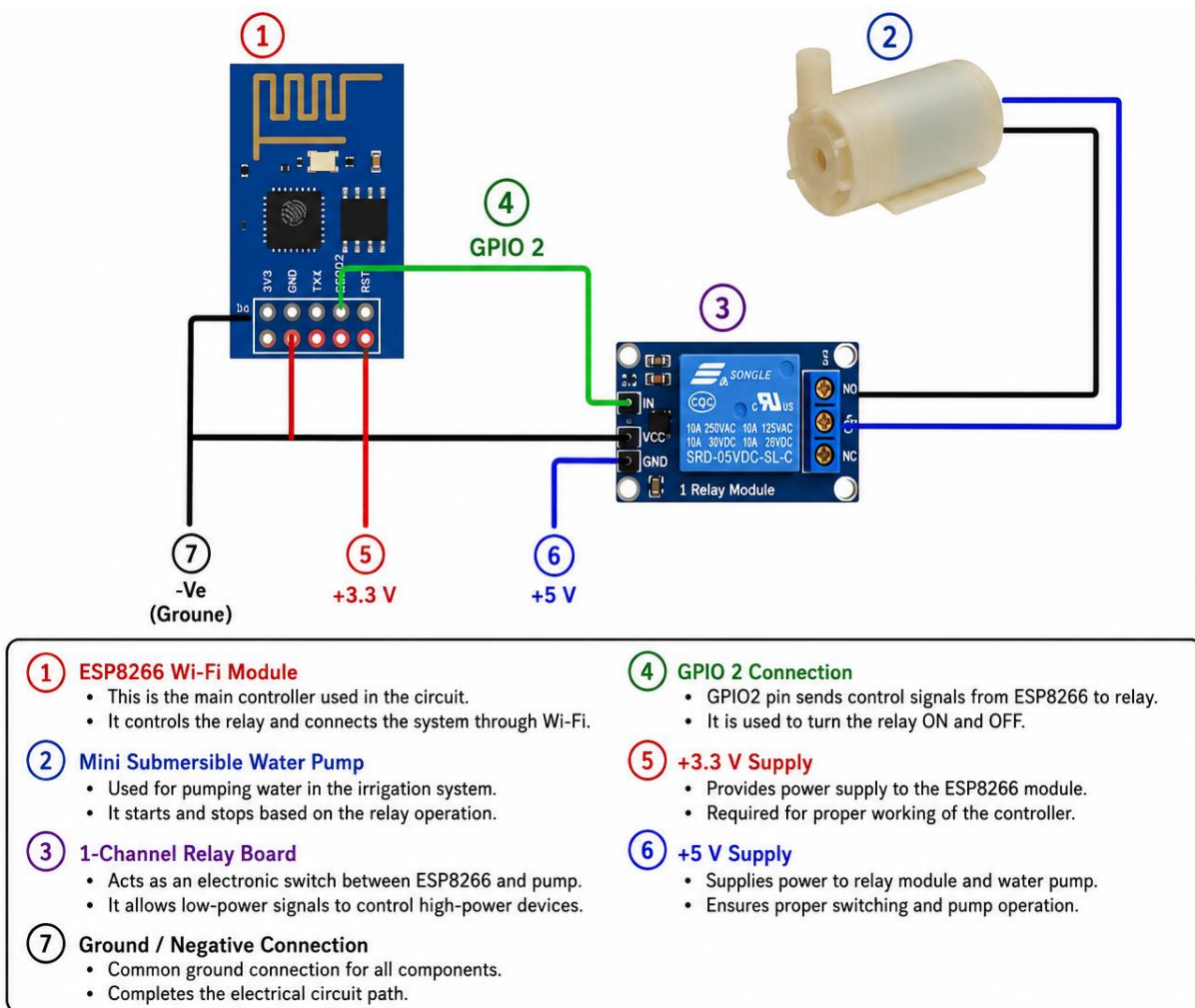


Figure 1.1

CONNECTION PROCEDURE:

The connection procedure for this circuit involves interfacing an Esp. If Systems ESP8266 Wi-Fi module with a 1-channel relay board and a mini submersible water pump. The ESP8266 acts as the control unit, while the relay safely switches the pump ON and OFF. Proper wiring is important to ensure smooth and reliable operation of the system.

First, the ESP8266 module should be connected to a regulated 3.3-volt power supply. The VCC pin is connected to the positive terminal, and the GND pin is connected to the negative terminal of the supply. Using the correct voltage is essential because the ESP8266 is sensitive to higher voltages.

The relay module is then connected to a separate 5-volt power source. Its VCC pin receives +5 V, and the GND pin is connected to the common ground shared with the ESP8266. A common ground connection helps maintain proper communication between the controller and the relay module.

The GPIO2 pin of the ESP8266 is connected to the input pin of the relay board. This GPIO pin sends digital signals that activate or deactivate the relay. When the signal is received, the relay changes its switching position and controls the connected pump circuit.

The mini submersible water pump is connected through the relay's switching terminals. One wire from the pump is directly connected to the power source, while the second wire passes through the Normally Open terminal of the relay. When the relay is activated, the electrical circuit closes and the pump starts operating.

After completing all the wiring connections, the circuit should be checked carefully before powering it ON. Loose wires or incorrect connections may prevent the system from functioning properly. Uploading a test program to the ESP8266 helps confirm whether the relay and pump are responding correctly.

Precautions must be followed while working with this circuit. The ESP8266 should never be supplied with more than 3.3 V because excess voltage can permanently damage the module. Similarly, the relay module should receive only the recommended operating voltage for safe performance.

All wires and terminals should be properly insulated to prevent short circuits and accidental contact. Since the water pump operates in a moist environment, extra care must be taken to keep electrical connections dry and secure. Wet or exposed wires can create safety risks and damage the components.

The relay should also be used within its specified current rating. Connecting a pump that draws excessive current may overheat the relay and reduce its lifespan. Choosing a suitable relay module ensures efficient switching and long-term reliability of the system.

Regular inspection and maintenance improve the durability of the circuit. Checking for loose wires, corrosion, and damaged connections helps avoid unexpected failures. By following proper connection procedures and safety precautions, the automated pump control system can operate safely and effectively for various IoT and water management applications.

CONCLUSION:

The Smart Irrigation System using a Soil Moisture Sensor and the Blynk application was successfully designed, developed, and implemented to provide an efficient and automated solution for modern irrigation management.

The primary objective of the project was to reduce unnecessary water consumption and minimize manual intervention in agricultural irrigation processes. The system achieved this objective by continuously monitoring the moisture level of the soil and automatically controlling the water pump based on real-time soil conditions.

The soil moisture sensor plays a major role in detecting the amount of water present in the soil. Whenever the soil becomes dry and the moisture level falls below the predefined threshold value, the system automatically switches on the water pump to irrigate the field.

Similarly, when the soil moisture reaches the required level, the pump is turned off automatically. This automation helps in maintaining optimum soil moisture conditions for plant growth while preventing over-irrigation and water wastage.

The integration of the Blynk application adds an important IoT feature to the system by enabling remote monitoring and control through a smartphone.

Farmers or users can easily monitor soil moisture values, check the status of the motor, and control the irrigation system from anywhere using an internet connection. This improves convenience, saves time, and reduces the need for continuous physical monitoring of agricultural fields. The real-time notification and remote accessibility provided by the Blynk platform make the system more intelligent and user-friendly.

Another important advantage of this project is its simplicity and low cost. The system is built using easily available components such as a microcontroller, soil moisture sensor, relay module, water pump, and Wi-Fi-enabled IoT platform.

Due to its affordable nature and simple design, the system can be implemented in small-scale farms, gardens, greenhouses, and even household gardening applications. It also contributes to sustainable agriculture by promoting efficient water management practices.

The project demonstrates how IoT technology can be effectively applied in agriculture to improve productivity and resource.

Furthermore, the system can be enhanced in the future by integrating additional sensors such as temperature, humidity, rainfall, and water level sensors. Advanced features like data analytics, machine learning, and weather prediction can also be added to make the irrigation process more intelligent and adaptive.

In conclusion, the Smart Irrigation System using Soil Moisture Sensor and Blynk application is a reliable, efficient, and eco-friendly solution for modern agriculture.

JUSTIFICATION OF THIS PROJECT:

Agriculture plays a vital role in sustaining human life and supporting the economy. One of the most important factors affecting crop yield and plant health is proper irrigation.

Plants require an adequate and timely supply of water for healthy growth; however, traditional irrigation methods often depend on manual monitoring and control.

This approach not only requires continuous human effort but also leads to inefficient water usage, including over-irrigation or under-irrigation.

In many regions, water scarcity has become a serious concern, making it necessary to adopt smarter and more efficient irrigation techniques.

To address these challenges, the Smart Irrigation System using a Soil Moisture Sensor and Blynk application has been developed.

This system automates the irrigation process by continuously monitoring the moisture level of the soil using a sensor.

When the soil moisture drops below a certain threshold, the system automatically activates the water pump to irrigate the plants.

Once the required moisture level is achieved, the pump is switched off automatically. This ensures that water is supplied only when needed and in appropriate quantities, thereby preventing water wastage.

In addition to automation, the system also enables real-time monitoring through the Blynk application. This allows users to observe soil conditions and control the irrigation system remotely using a smartphone. Such a feature enhances convenience, especially for farmers who manage large agricultural fields or are unable to be physically present at all times.

The proposed system is cost-effective, simple to design, and easy to implement using commonly available components. It is suitable for various applications such as home gardens, agricultural fields, greenhouse farming, and smart farming systems.

By improving water efficiency and reducing manual labour, this system contributes significantly to modern agricultural practices.

Therefore, this project is justified as it provides an efficient, automated, and sustainable solution for irrigation management. It helps conserve water resources, reduces human effort, and supports the advancement of smart agriculture technologies, which are essential for future farming needs.

The system helps reduce labour costs by minimizing the need for manual field inspection and irrigation control, making farming more efficient and less time-consuming. It improves crop yield quality by maintaining optimal soil moisture levels consistently, which prevents plant stress caused by overwatering or underwatering.

APPENDIX:

An irrigation control dashboard created using the Blynk mobile application for monitoring and controlling an automated watering system.

The gauge widget (V0) is used to display sensor readings such as soil moisture values, while the buttons V1 and V2 are virtual controls used to operate devices like water pumps or valves remotely.

The dashboard provides a simple IoT interface that allows users to monitor irrigation conditions and control the system wirelessly through a smartphone. Fig. 1.2 shows the Blynk IoT application interface used for monitoring soil moisture values and controlling the irrigation system remotely.

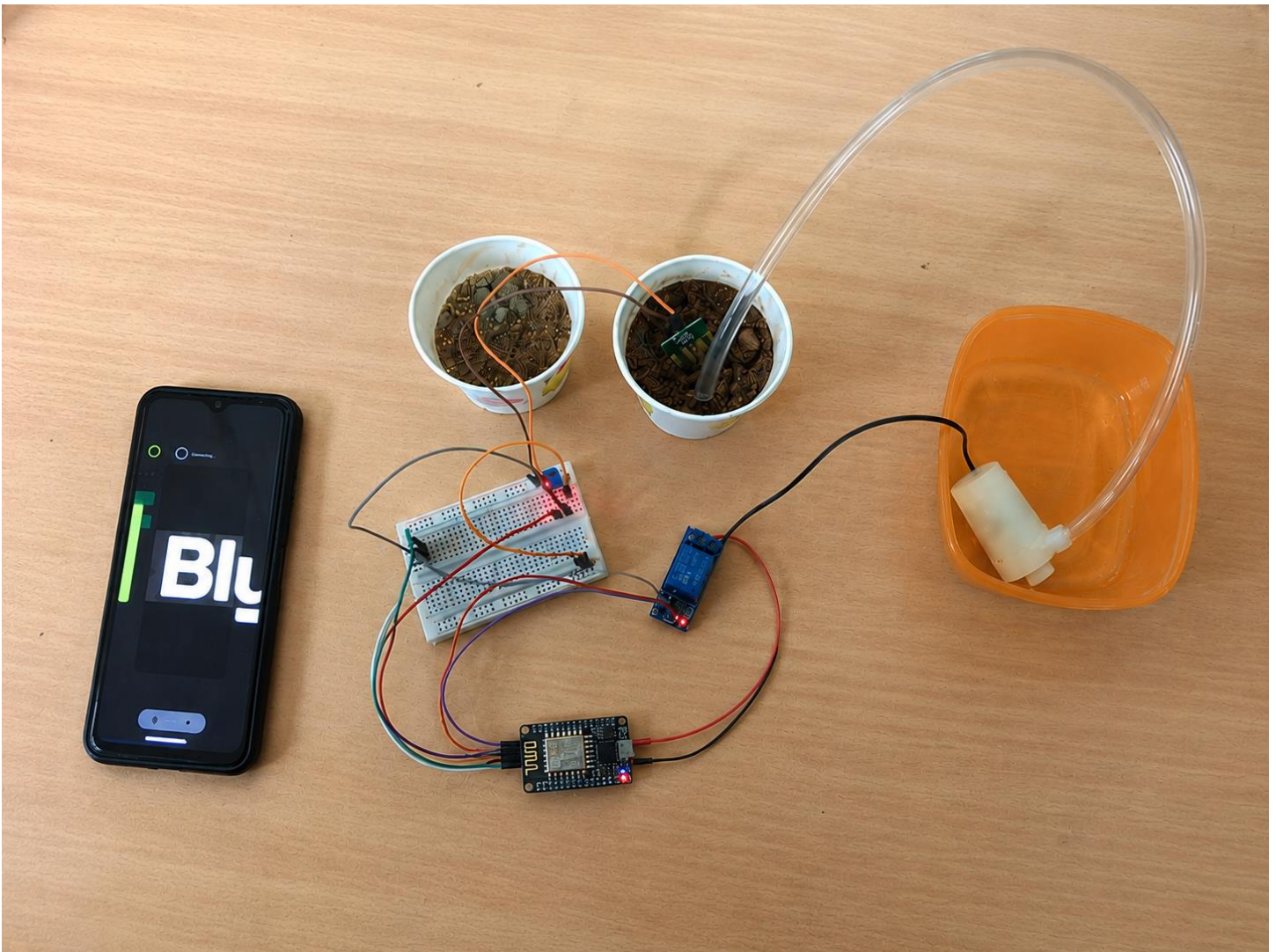


Figure 1.2

BLYNK DASHBOARD:

The below fig 1.3 shows the Blynk IoT application interface used in the smart irrigation system project. It displays the soil moisture sensor values in real time using a gauge meter.

When the moisture value is high (866), the pump status becomes ON, indicating dry soil condition. When the moisture value is low (590), the pump status becomes OFF, indicating sufficient water content in the soil. Thus, the application helps in automatic monitoring and control of irrigation through mobile connectivity.

The Blynk application is connected with the Node MCU ESP8266 through Wi-Fi communication. It allows the user to monitor the soil condition from anywhere using a smartphone. The ON and OFF indication shown in the app represents the working condition of the relay and water pump. This reduces manual effort and ensures efficient water management in agricultural applications.

The system continuously checks the moisture level of the soil and updates the readings in the mobile application instantly. Whenever the soil becomes dry, the pump is automatically activated to supply water to the plant.

After reaching the required moisture level, the pump is turned OFF automatically to avoid excess watering. Hence, the smart irrigation system provides an efficient, low-cost, and automated solution for modern farming practices.

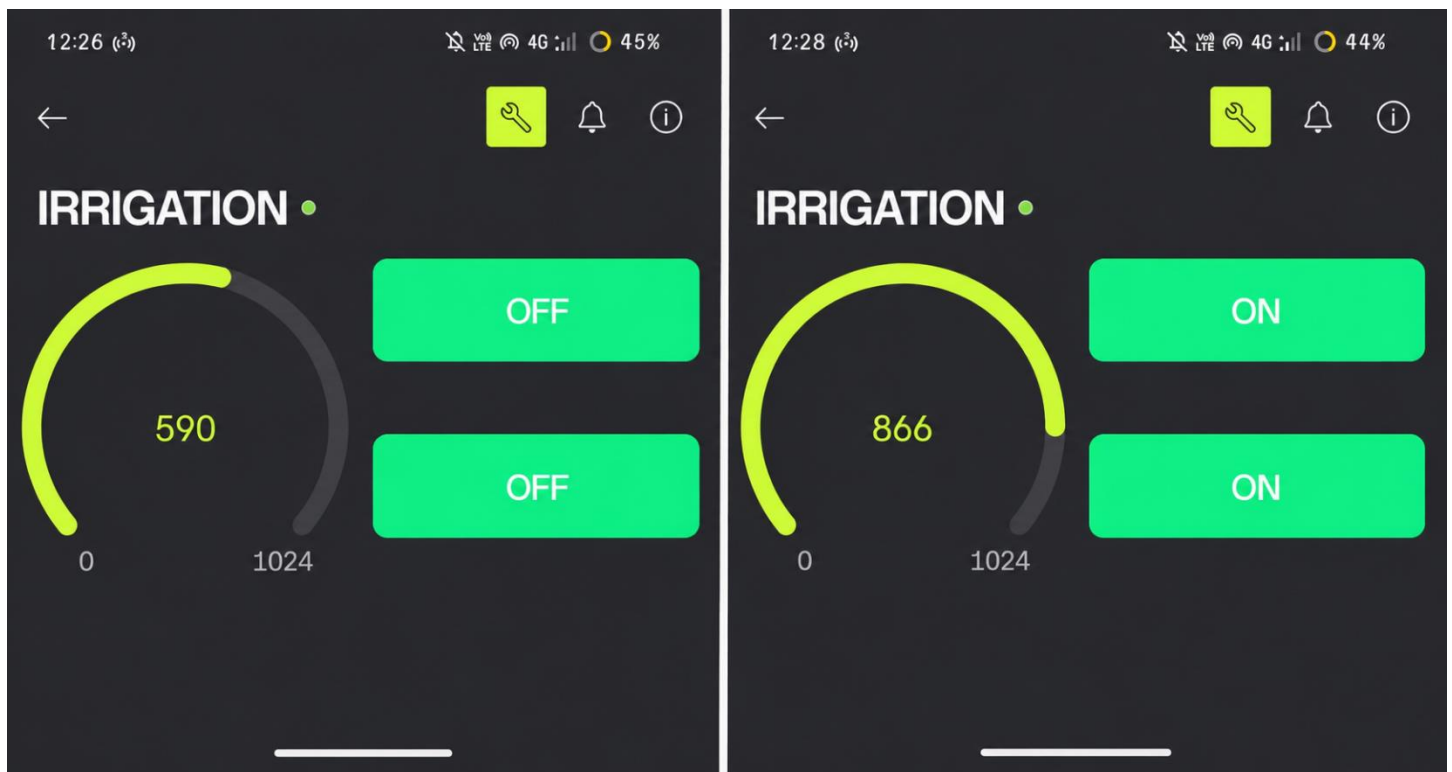


Figure 1.3

ACKNOWLEDGEMENT:

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of this project on the smart irrigation system using the Esp. if Systems ESP8266 module.

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